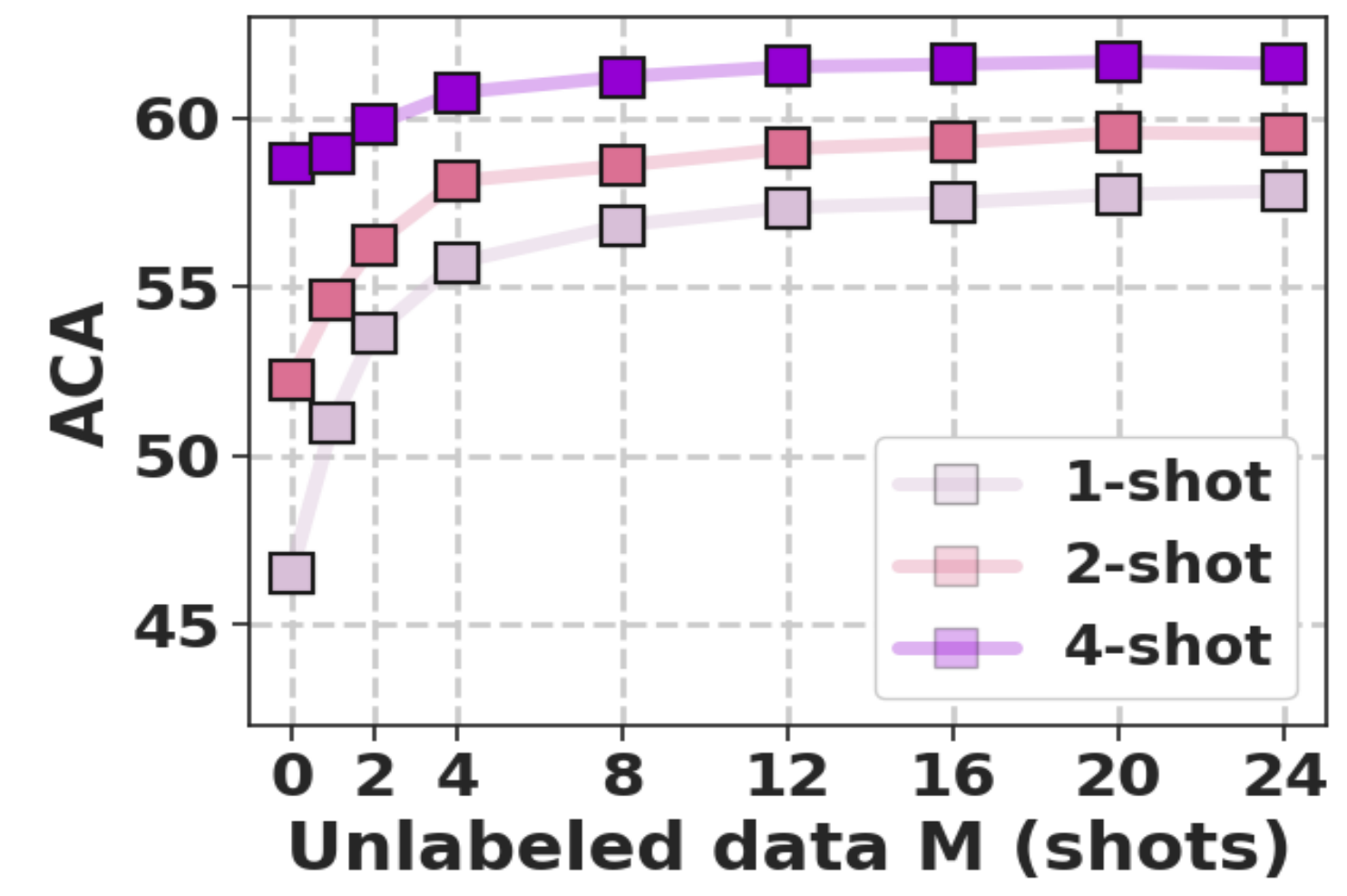
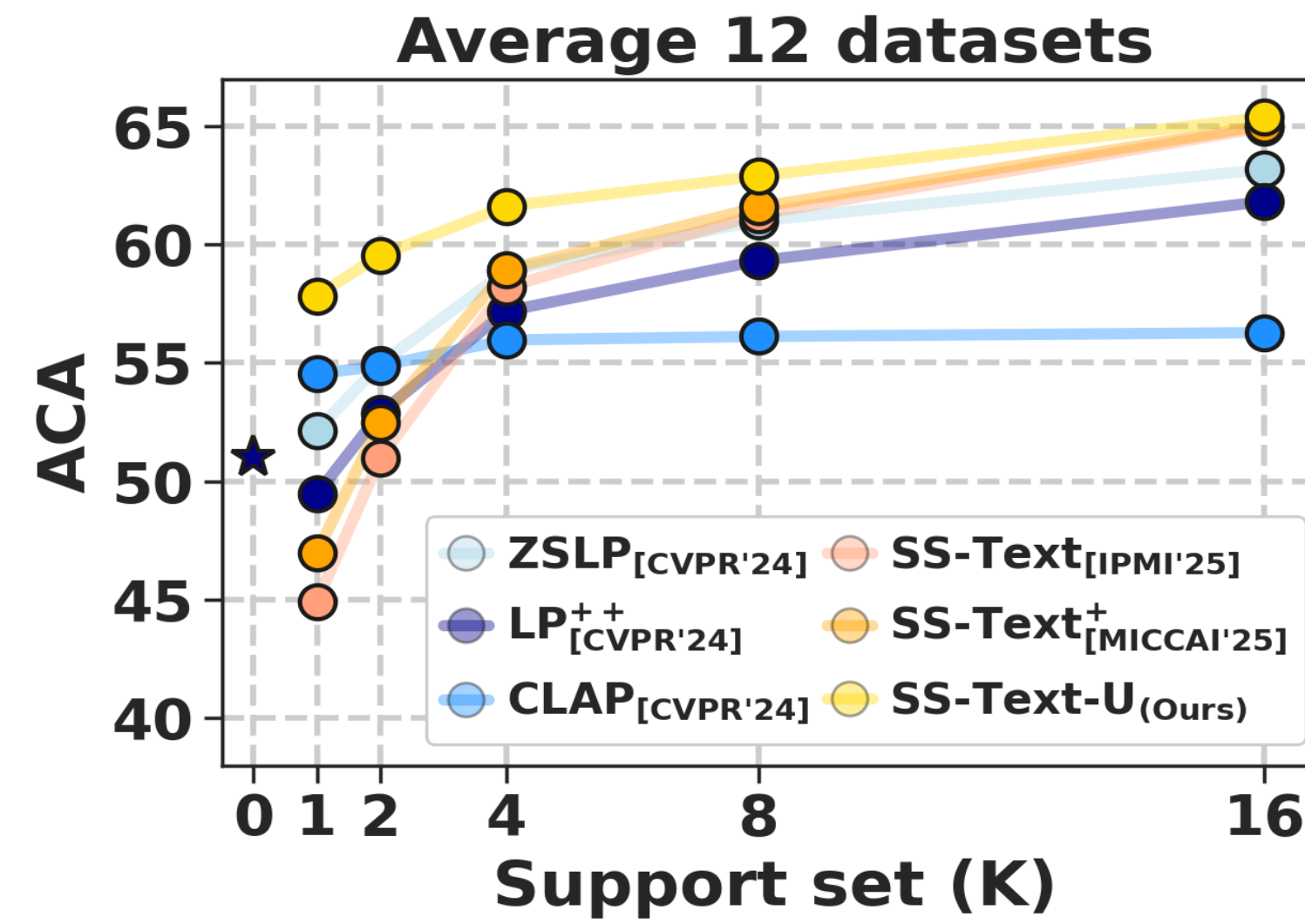
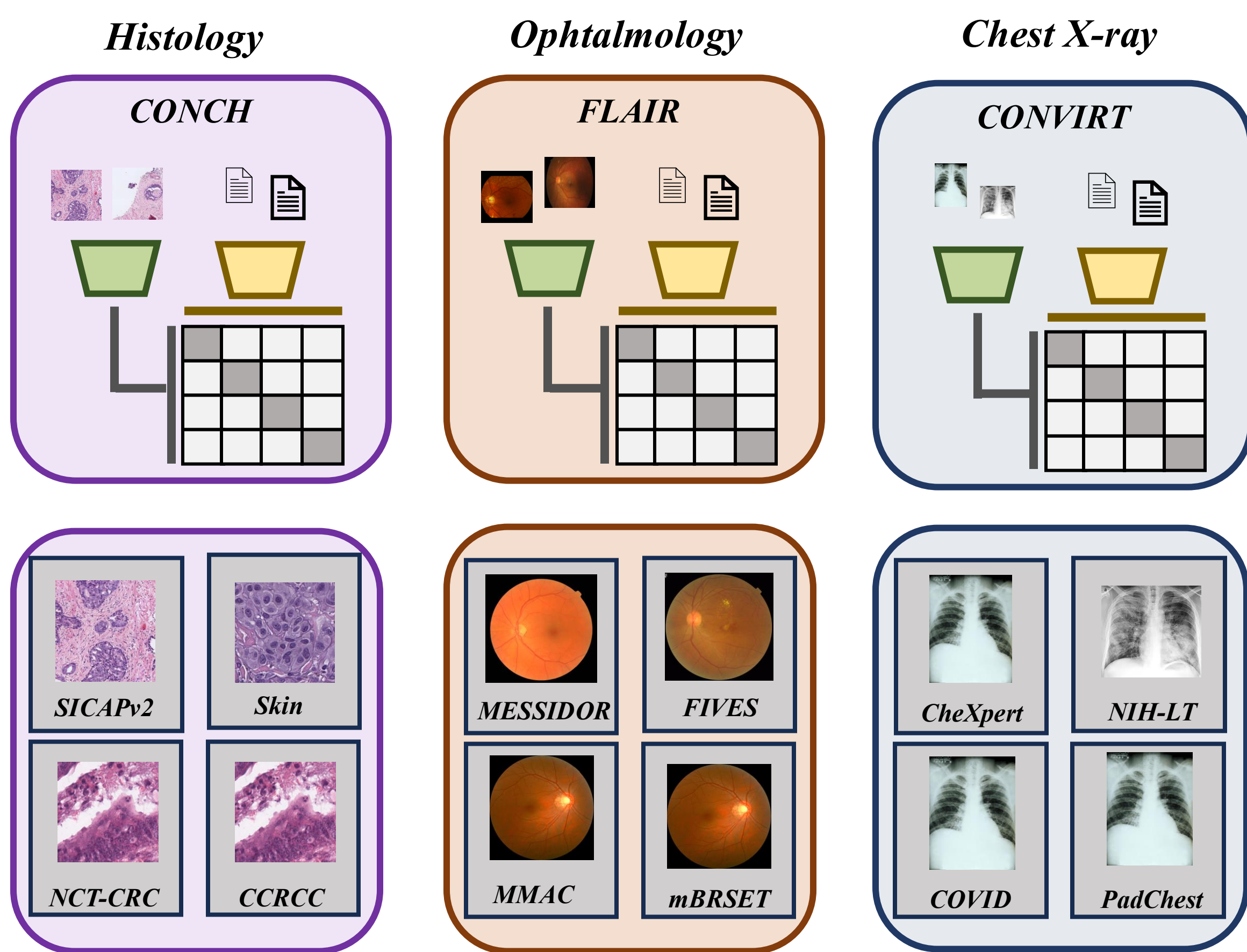


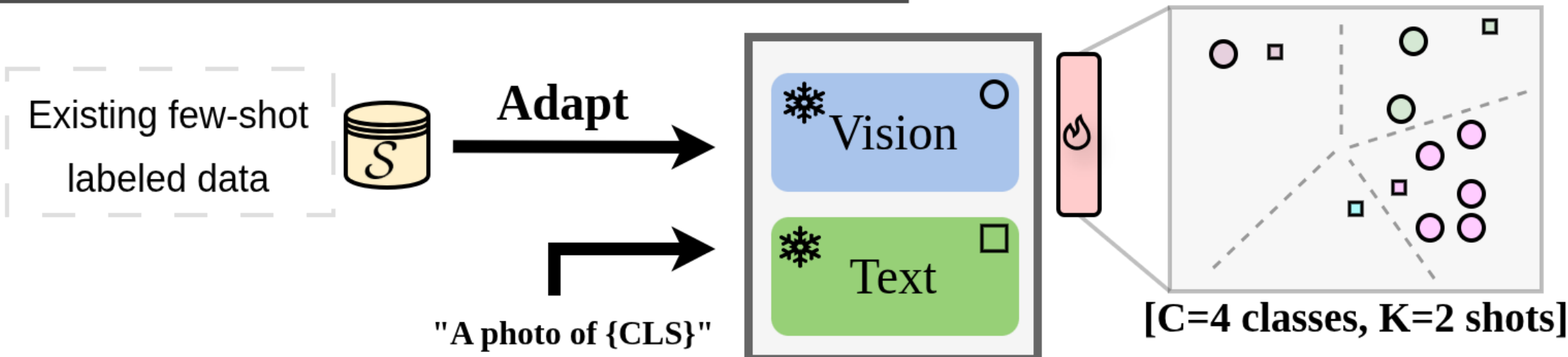
- **VLM embeddings** can be **efficiently transferred** using **few-shot** annotated examples.
- **Current literature assumes incomplete scenarios**, where annotated data is suddenly available, without considering **access to the unlabeled data pool before costly expert annotation**.
- We extend existing few-shot classification setting to the **semi-supervised scenario**, and propose a novel efficient block-wise optimizer that **reduces annotation effort by more than 50%**.



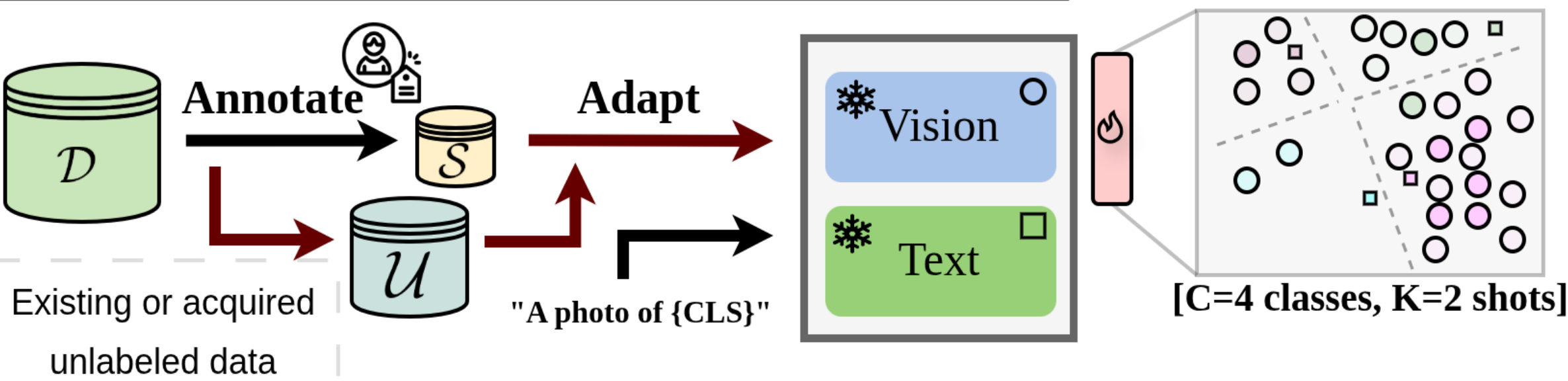
Why semi-supervised learning?

In many scenarios, unlabeled data is available or can be acquired, while the bottleneck lies in expert-annotation of such data.

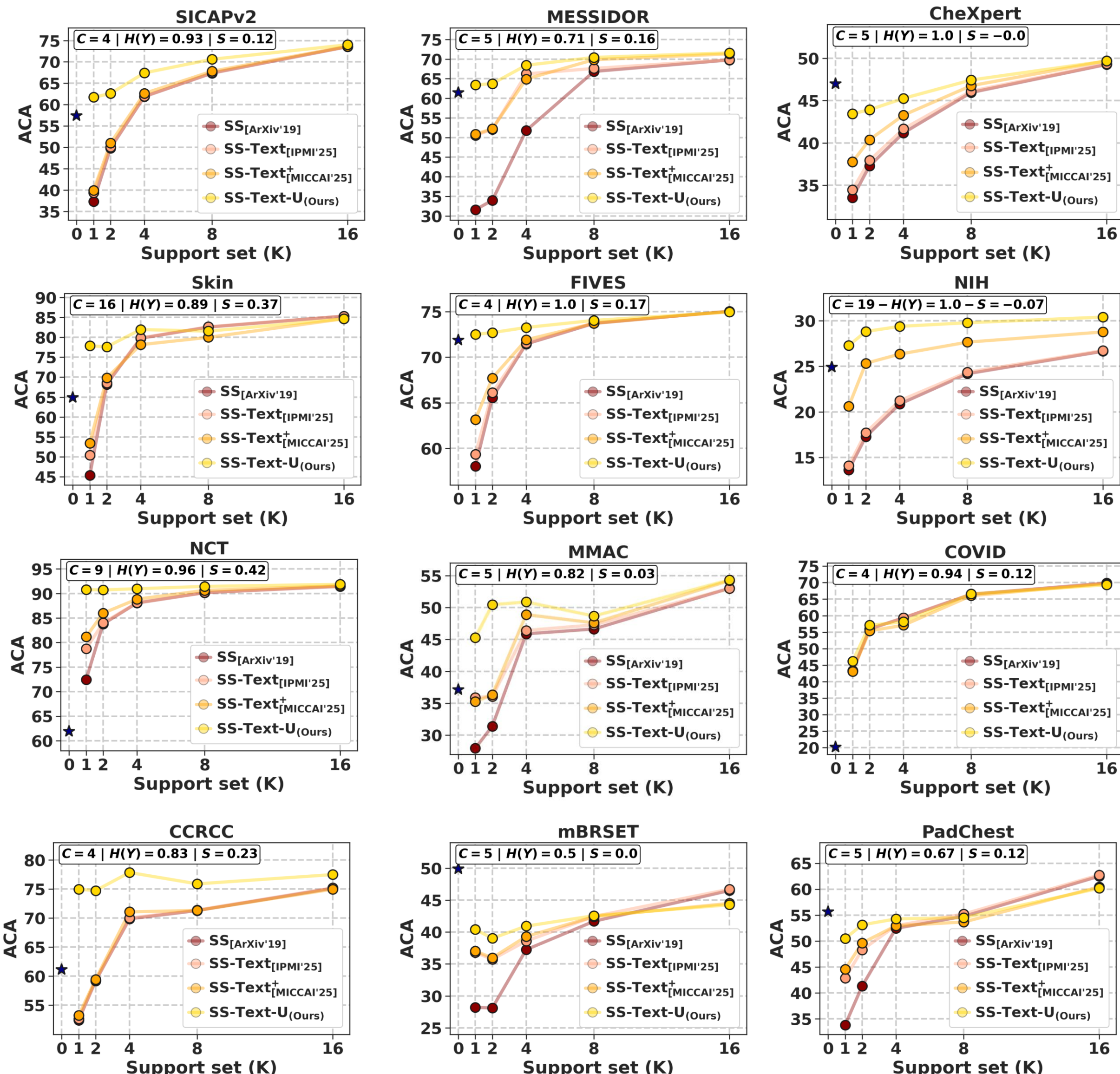
Standard few-shot adaptation scenario.



Semi-supervised few-shot adaptation (Ours).



Detailed per-dataset results



Improving low-shot adaptation with unlabelled data

1 — Closed-form text-constrained linear probe solution.

$$\min_{\mathbf{W}} \mathcal{L}(\mathbf{W}) = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \mathcal{H}(y_i, \mathbf{p}_i(\mathbf{W})) + \sum_{c \in \mathcal{Y}} \lambda_c^T \|\mathbf{w}_c - \mathbf{t}_c\|_2^2$$

Cross-entropy loss:

$$\mathcal{H}(y_i, \mathbf{p}_i(\mathbf{W})) = - \sum_{c \in \mathcal{Y}} y_{i,c} \ln(p_{i,c}(\mathbf{W}))$$

SoftMax similarity:

$$p_{i,c}(\mathbf{W}) = \frac{\exp(\mathbf{v}_i^T \mathbf{w}_c / \tau)}{\sum_{j \in \mathcal{Y}} \exp(\mathbf{v}_i^T \mathbf{w}_j / \tau)}$$

$$\mathcal{H}(y_i, \mathbf{p}_i(\mathbf{W})) = - \sum_{c \in \mathcal{Y}} y_{i,c} (\mathbf{v}_i^T \mathbf{w}_c / \tau) + \sum_{c \in \mathcal{Y}} y_{i,c} \ln \left(\sum_{j \in \mathcal{Y}} \exp(\mathbf{v}_i^T \mathbf{w}_j / \tau) \right)$$

$\mathcal{H}^t(y_i, \mathbf{W})$ $\mathcal{H}^{\text{cont}}(\mathbf{W})$

$$\mathbf{w}_c^t = \arg \min_{\mathbf{w}_c} \frac{\partial \mathcal{L}_{\text{FEW-SHOT}}(\mathbf{W})}{\partial \mathbf{w}_c} = \frac{1}{2\lambda_c^T |\mathcal{S}| \tau} \sum_{i \in \mathcal{S}} y_{i,c} \mathbf{v}_i + \mathbf{t}_c$$

2 — Extension using unsupervised data.

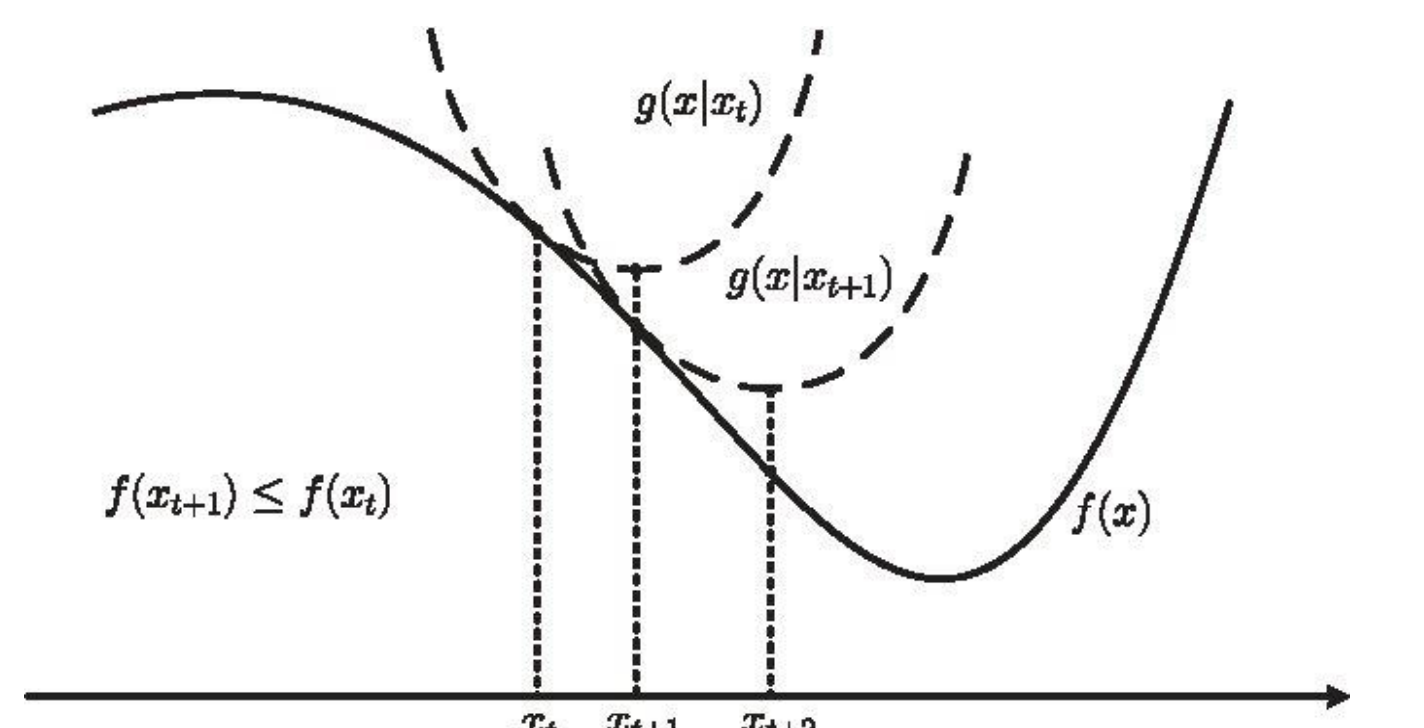
$$\min_{\mathbf{W}, \mathbf{z}} \mathcal{L}_{\text{SEMI}}(\mathbf{W}, \mathbf{z}) = \mathcal{L}_{\text{FEW-SHOT}}(\mathbf{W}) + \lambda^U \mathcal{L}_U(\mathbf{W}, \mathbf{z})$$

Unsupervised data learning objective:

$$\mathcal{L}_U(\mathbf{W}, \mathbf{z}) = \frac{1}{|\mathcal{U}|} \sum_{i \in \mathcal{U}} \mathcal{H}^t(\mathbf{z}_i, \mathbf{W}) \text{ s.t. } \hat{\mathbf{m}} = \mathbf{m}$$

Label frequency
(estimated from labeled set)

Pseudo-labels are treated as regularized learnable parameters.



Block-wise efficient optimizer.

For t in T :

Step 1: Z-block update.

$$\min_{\mathbf{z}} \mathcal{L}_U(\mathbf{W}^{t-1}, \mathbf{z}) = - \frac{1}{|\mathcal{U}|} \sum_{i \in \mathcal{U}} \sum_{c \in \mathcal{Y}} \mathbf{z}_{i,c} (\mathbf{v}_i^T \mathbf{w}_c^{t-1} / \tau) \text{ s.t. } \hat{\mathbf{m}} = \mathbf{m}$$

$$\mathbf{Q}^* = \text{Diag}(\mathbf{r}^{(T_{\text{OT}})}) \mathbf{Q}^{(0)} \text{Diag}(\mathbf{c}^{(T_{\text{OT}})}),$$

$$\mathbf{r}^{(t_{\text{OT}})} = \mathbf{m} / (\mathbf{Q}^{(0)} \mathbf{c}^{(t_{\text{OT}}-1)}), \mathbf{c}^{(t_{\text{OT}})} = \mathbf{u}_{(|\mathcal{U}|)} / (\mathbf{Q}^{(0)} \mathbf{r}^{(t_{\text{OT}})})$$

Solved as an Optimal Transport problem using regularized Sinkhorn algorithm.

Step 2: W-update.

$$\mathbf{w}_c^t = \arg \min_{\mathbf{w}_c} \frac{\partial \mathcal{L}_{\text{SEMI}}(\mathbf{W}, \mathbf{z}^t)}{\partial \mathbf{w}_c} = \frac{1}{2\lambda_c^T |\mathcal{S}| \tau} \sum_{i \in \mathcal{S}} y_{i,c} \mathbf{v}_i + \frac{\lambda_c^U}{2\lambda_c^T |\mathcal{U}| \tau} \sum_{i \in \mathcal{U}} \mathbf{z}_{i,c}^t \mathbf{v}_i + \mathbf{t}_c$$